AI PROJECT – GAME SIMULATOR

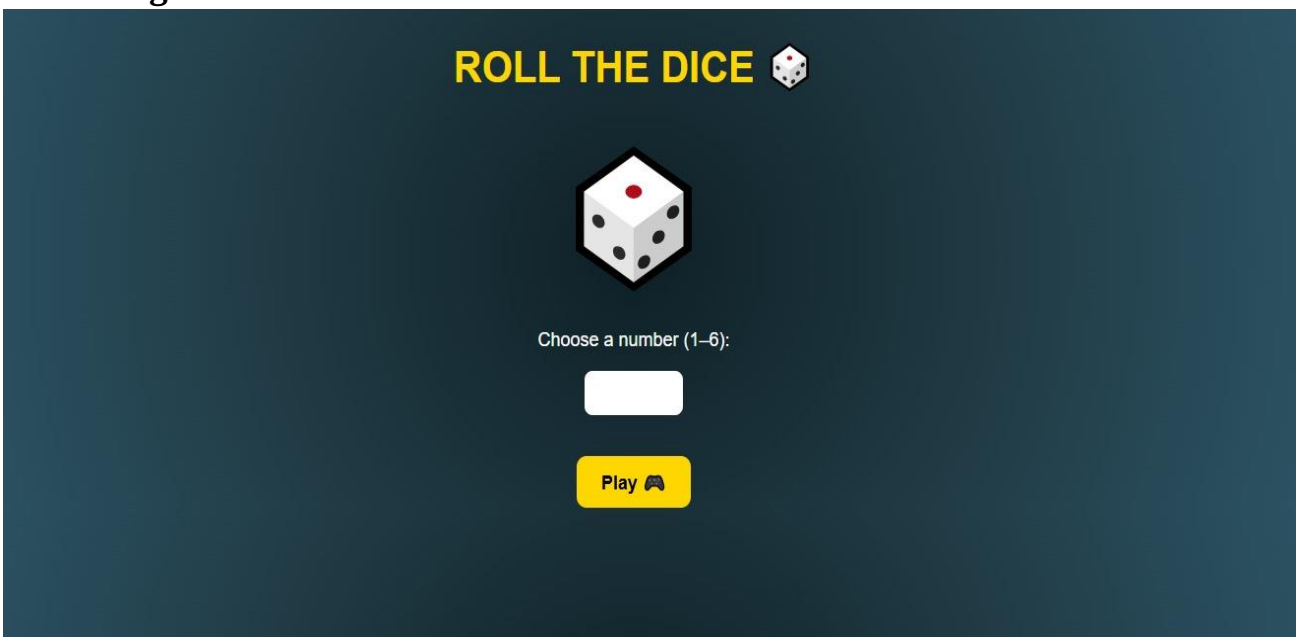
Project Overview

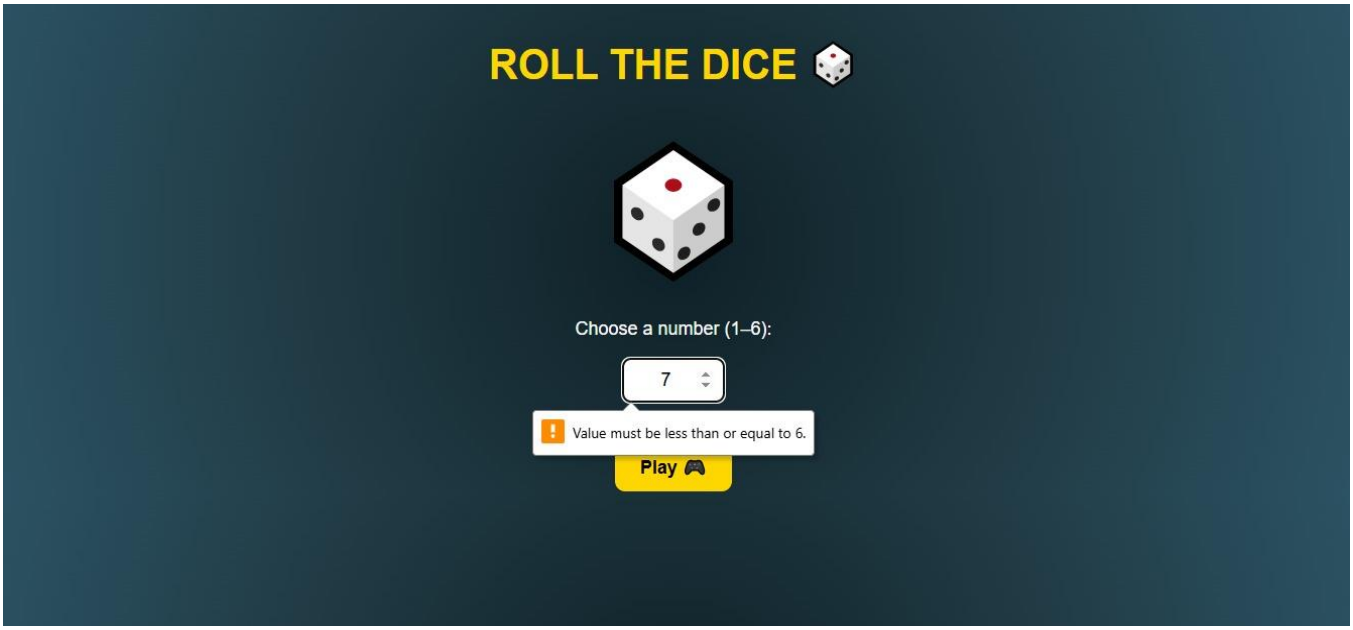
This project is a simple web-based dice game built using Python and Django.
The application allows a user to:

1. Choose a number between 1 and 6
2. Simulate a dice roll using a Random Number Generator
3. Determine whether a user wins or loses based on the matching number

How to run the project (Screenshots)

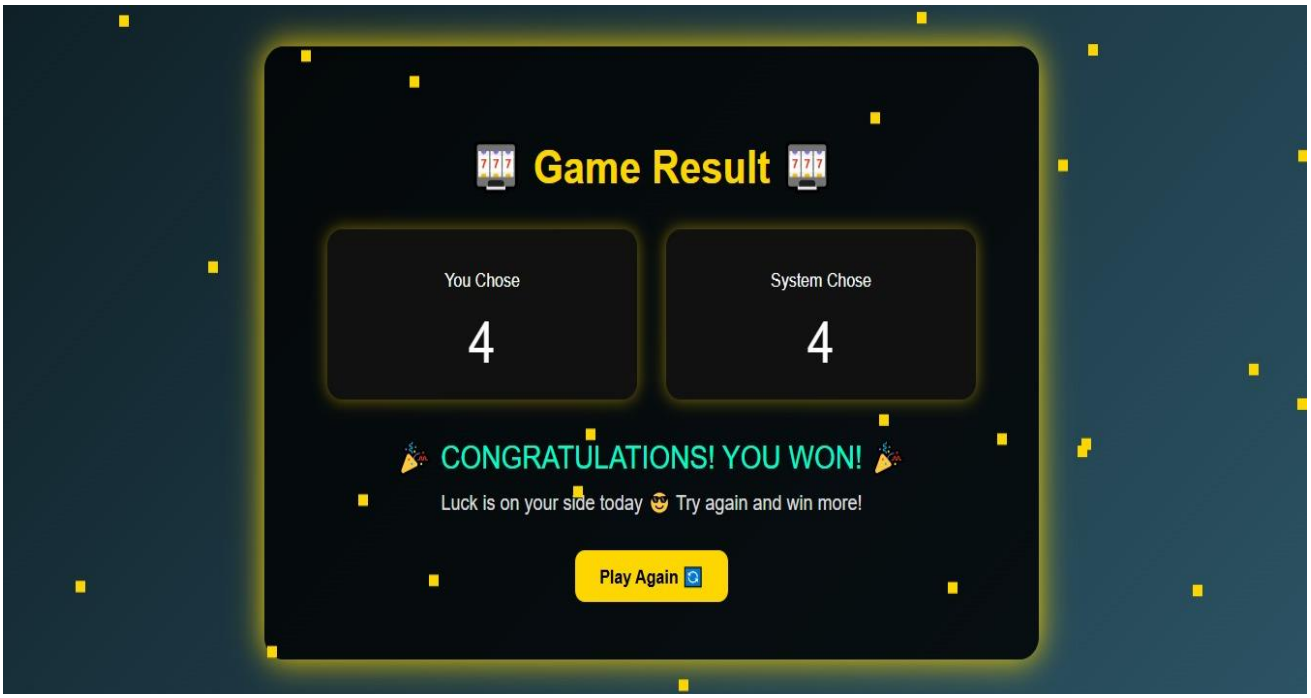
Home Page



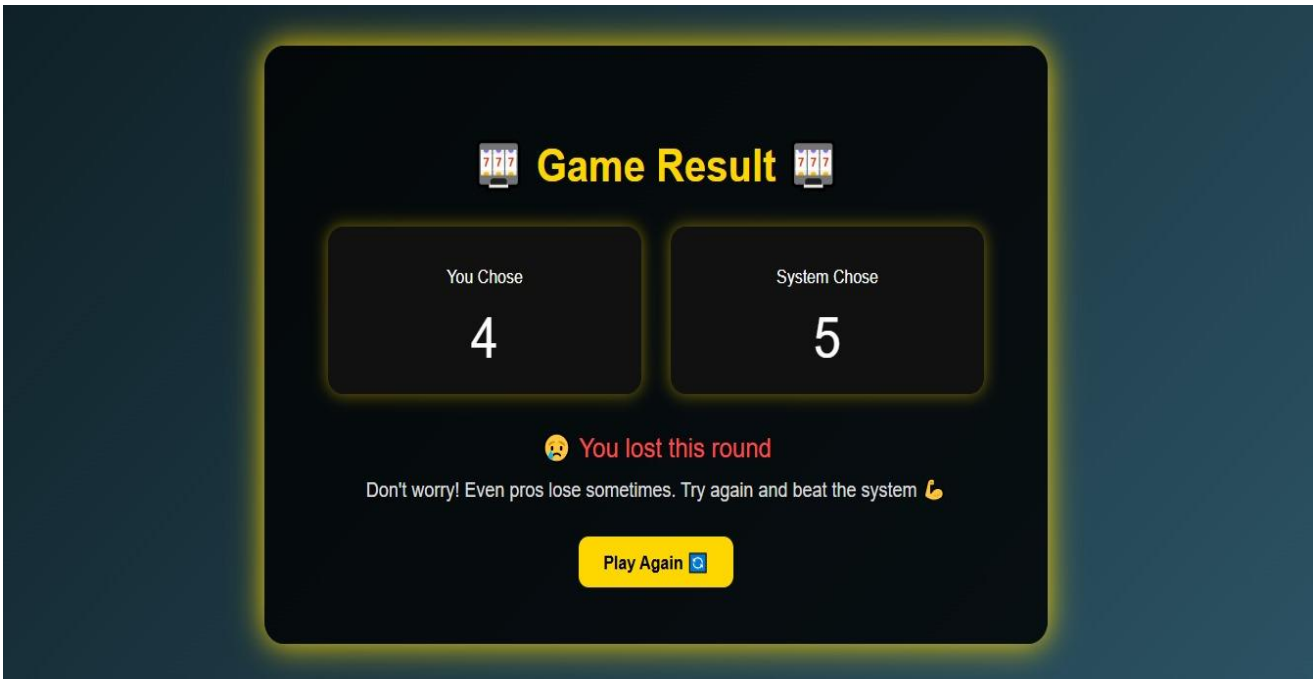


Result Page

Result- Win



Result-Lose



Objectives

To build a functional Django web application with the help of AI
To demonstrate randomness using a Random Number Generator
To implement form handling and validation with the use of AI prompts
To create an interactive and visually appealing user interface
To document the development process using AI assistance

Technologies Used

Python, Django, HTML, CSS, and JavaScript, Claude (claude-sonnet-4-6), OpenAI GPT-4o, Gemini 1.5 Pro — Google

System Architecture

Architecture Type

Client-Server Architecture

Flow

User interacts with the frontend
Request sent to Django Backend
Backend processes logic
Response returned to the frontend

Project Structure

AI/
settings.py
urls.py
game/
views.py
urls.py
templates/
home.html
result.html

Implementation Steps

Step 1: Project Setup

Django installation
Created a new Django project (AI)
Created an app (game)
Registered the app in settings.py

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>cd One DRIVE
The system cannot find the path specified.

C:\Users\hp>cd OneDrive

C:\Users\hp\OneDrive>cd Desktop

C:\Users\hp\OneDrive\Desktop>cd AI PROJECT

C:\Users\hp\OneDrive\Desktop\AI PROJECT>python -m venv myvenv

C:\Users\hp\OneDrive\Desktop\AI PROJECT>.\myvenv\Scripts\activate

(myvenv) C:\Users\hp\OneDrive\Desktop\AI PROJECT>pip install django
Collecting django
  Downloading django-6.0.3-py3-none-any.whl.metadata (3.9 kB)
Collecting asgiref>=3.9.1 (from django)
  Downloading asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.5.0 (from django)
  Downloading sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Collecting tzdata (from django)
  Downloading tzdata-2025.3-py2.py3-none-any.whl.metadata (1.4 kB)
Downloading django-6.0.3-py3-none-any.whl (8.4 MB)
   ---------------------------------------- 8.4/8.4 MB 1.2 MB/s eta 0:00:00
Downloading asgiref-3.11.1-py3-none-any.whl (24 kB)
Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
Downloading tzdata-2025.3-py2.py3-none-any.whl (348 kB)
```

Step 2: URL Configuration

Created a urls,py file in the app (game) where I created routes for the home page and the /play/ (Game logic)
Connected app URLs to the main project (AI)

Step 3: Creating Views

Home View

Display the home page
Contains the input form for user interaction

Play View(Core Logic)

User submits a number via the form
Input is retrieved using a request.POST.get()
Input is validated:
  Must not be empty
  Must be between 1-6
The system generates a random number
Numbers are compared
Results are displayed

Step 4: Frontend Development

Home Page

Input field restricted to numbers (1-6)
Play button
Error handling display
Animated rolling dice
Casino-themed styling

Result Page

Display:
 User number
 System number
 Result
Includes:
  Confetti animation on win
  Encouraging message on loss
Styled with a modern casino interface

Step 5: Testing

Test Case	Input	Expected Result	Outcome
Valid Input	3	Win/Lose	Passed

Empty Input	None	Error message	Passed
Out of Range	10	Error message	Passed

Challenges and Errors Encountered

1. Missing CSRF Token

Prompt Template Used

I need help understanding this error message from my Python/Django application.

Here's the complete error message and stack trace:

Forbidden (403)
CSRF verification failed. Request aborted.

Help

Reason given for failure:
CSRF token missing.

In general, this can occur when there is a genuine Cross Site Request Forgery, or when Django's CSRF mechanism has not been used correctly. For POST forms, you need to ensure:

- Your browser is accepting cookies.
- The view function passes a request to the template's render method.
- In the template, there is a {% csrf_token %} template tag inside each POST form that targets an internal URL.
- If you are not using CsrfViewMiddleware, then you must use csrf_protect on any views that use the csrf_token template tag, as well as those that accept the POST data.
- The form has a valid CSRF token. After logging in in another browser tab or hitting the back button after a login, you may need to reload the page with the form, because the token is rotated after a login.

You're seeing the help section of this page because you have DEBUG = True in your Django settings file. Change that to False, and only the initial error message will be displayed.
You can customize this page using the CSRF_FAILURE_VIEW setting.

My application context:

- This happened when I was trying to test my application on the UI

- The application is a game that uses a Random Number Generator to select a number randomly from 1-6 and compare it to the chosen number by the user to determine if the user wins or loses the game.
- I'm using Python 3.13.2 and Django 26.0.1

Could you:

1. Explain what this error means in simple, non-technical terms
2. Identify the most relevant lines in the stack trace (which ones actually point to my code)
3. List 2-3 of the most likely causes based on this type of error
4. Suggest what specific information I should look for in my code
5. Provide a step-by-step debugging approach I could follow

I'm particularly unfamiliar with Cross-Site Request Forgery mentioned in the error, so an extra explanation there would help.

Fix:
Django expected a security token but didn’t receive it, so it blocked the request. I added {% crsf_token %} inside my form

2. URL Not Found (404)
Prompt Template

I need help diagnosing the root cause of an error in my Python/Django application.

Here's the error and stack trace:
Traceback (most recent call last):

  File "C:\Users\hp\AppData\Local\Programs\Python\Python313\Lib\threading.py", line 1041, in _bootstrap_inner

    self.run()

    ~~~~~~~~^^^

  File "C:\Users\hp\AppData\Local\Programs\Python\Python313\Lib\threading.py", line 992, in run

    self._target(*self._args, **self._kwargs)

    ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```
  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\utils\autoreload.py", line 64, in wrapper

    fn(*args, **kwargs)

    ~~^^^^^^^^^^^^^^^^^^

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\core\management\commands\runserver.py", line 134, in inner_run

    self.check(**check_kwargs)

    ~~~~~~~~~~^^^^^^^^^^^^^^^^^

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\core\management\base.py", line 496, in check

    all_issues = checks.run_checks(

        app_configs=app_configs,

    ...<2 lines>...

        databases=databases,

    )

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\core\checks\registry.py", line 89, in run_checks

    new_errors = check(app_configs=app_configs, databases=databases)

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\core\checks\urls.py", line 44, in check_url_namespaces_unique

    all_namespaces = _load_all_namespaces(resolver)

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\core\checks\urls.py", line 63, in _load_all_namespaces

    url_patterns = getattr(resolver, "url_patterns", [])

  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\utils\functional.py", line 47, in __get__
```

```
    res = instance.__dict__[self.name] = self.func(instance)
                                         ~~~~~~~~~~^^^^^^^^^^
  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\urls\resolvers.py", line 729, in url_patterns
    patterns = getattr(self.urlconf_module, "urlpatterns", self.urlconf_module)
                       ^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\utils\functional.py", line 47, in __get__
    res = instance.__dict__[self.name] = self.func(instance)
                                         ~~~~~~~~~~^^^^^^^^^^
  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\myvenv\Lib\site-packages\django\urls\resolvers.py", line 722, in urlconf_module
    return import_module(self.urlconf_name)
  File "C:\Users\hp\AppData\Local\Programs\Python\Python313\Lib\importlib\__init__.py", line 88, in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "<frozen importlib._bootstrap>", line 1387, in _gcd_import
  File "<frozen importlib._bootstrap>", line 1360, in _find_and_load
  File "<frozen importlib._bootstrap>", line 1331, in _find_and_load_unlocked
  File "<frozen importlib._bootstrap>", line 935, in _load_unlocked
  File "<frozen importlib._bootstrap_external>", line 1022, in exec_module
  File "<frozen importlib._bootstrap_external>", line 1160, in get_code
  File "<frozen importlib._bootstrap_external>", line 1090, in source_to_code
  File "<frozen importlib._bootstrap>", line 488, in _call_with_frames_removed
```

```
  File "C:\Users\hp\OneDrive\Desktop\AI PROJECT\AI\AI\urls.py", line 6

    path("include('game.urls'))

        ^^^^^^^^^^^^^^^^^^^^^^^
```

SyntaxError: invalid syntax. Perhaps you forgot a comma?

Here are the relevant code snippets from files mentioned in the stack trace:

File: (AI) urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path(" include('game.urls'))
]
```

Additional context:

- This error occurs when I try to access my UI
- It happens consistently
- I've already tried include commas in my code immediately after the single quotes and at the end of the last url pattern.

Could you please:

1. Identify the likely root cause (not just the symptoms)
2. Explain the chain of events leading to this error
3. Suggest 2-3 specific code changes that might fix the issue
4. Recommend tests I could write to verify the fix
5. Explain any patterns or anti-patterns you notice that might cause similar problems elsewhere
6. Suggest debugging tools or techniques specific to this type of error

I'm most confused about new_errors = check(app_configs=app_configs, databases=databases) mentioned in the error message so detailed explanation there would be helpful.

Fix:
I added the necessary commas and properly linked the project's urls.py and the app's urls.py.

Use of AI in the Project

1. To Understand How The Random Number Generator works and how to go about it

Prompt Template Used

I am a junior developer with basic knowledge of Python/Django, I can set up a Django project, models, different views, apply migrations, register models in the admin, create a SuperUser, enable static files in Django and create static and media directories inside the project, implement Http methods used in views, and also create different HTML templates with CSS stylings, but I do not understand the Random Number Generator and how to go about it. Take the role of a Senior developer and explain the RNG concept and how to go about it, relating it to my already known concepts from above for better understanding and also ask me questions that will determine my understanding of the concept and give examples of code snippets to go about the RNG.

Outcomes:

Understood the use of a Random Number Generator(RNG)
RNG concept explained using Django Knowledge(request- processing- response cycle)
Understood how RNG works in Python
Understood how to connect RNG to my existing skills
Got edge cases to consider when implementing the RNG in my project
Got self-assessment questions that improved my understanding

2. Documentation(Include Comprehensive Code Comments in my view)

Prompt Used

Please create comprehensive code comments for this function, following Python conventions

```
def play(request):
    if request.method == "POST":
        user_number = request.POST.get('number')

        if not user_number:
            return render(request, 'home.html', {
                'error': 'Please enter a number between 1 and 6'
            })

        user_number = int(user_number)

        if user_number < 1 or user_number > 6:
            return render(request, 'home.html', {
                'error': 'Number must be between 1 and 6'
            })

        system_number = random.randint(1, 6)

        if user_number == system_number:
            result = "You Win! 🎉"
        else:
            result = "You Lose 😞"

        return render(request, 'result.html', {
            'user_number': user_number,
            'system_number': system_number,
            'result': result
        })

    return render(request, 'home.html')
```

The documentation should include:

A clear description of what the function does

All parameters with types and descriptions
Return value with type and description
Any exceptions or errors that might be thrown
Example usage
Any important notes or edge cases that developers should be aware of
Outcomes:
Generated structured comments
Improved Code readability
Helped document parameters, returns, and logic

3. Documentation(Explaining the intent and Logic behind a code)

I need help documenting the intent and logic behind this code. Please:

```
  <div class="dice">🎲</div>

  <div class="container">

    {% if error %}
      <p class="error">{{ error }}</p>
    {% endif %}

    <form method="POST" action="{% url 'play' %}">
      {% csrf_token %}

      <label>Choose a number (1–6):</label><br><br>

      <input type="number" name="number" min="1" max="6" required>

      <br><br>

      <button type="submit">Play 🎮</button>
    </form>

  </div>

</body>
```

Explain what this code is trying to accomplish at a high level

Break down the logic step-by-step

Identify any assumptions or edge cases in the implementation

Suggest inline comments for complex parts

Note any potential improvements while maintaining the original functionality

Outcomes:

Helped in documenting a high-level explanation for the code
Helped in the documentation of the step-by-step logic breakdown
Generated assumptions and edge cases
Suggested inline code comments
Suggested potential improvements without changing the functionality

Conclusion

This project demonstrates how a simple game can be built using Django while incorporating randomness, user interaction, and frontend design. It highlights the importance of validation, security, and structured development in building reliable web applications.

Additionally, the successful implementation of this project was significantly enhanced by well-structured, effective AI prompts. These prompts supported a deeper understanding of key concepts such as the Random Number Generator, improved debugging processes, and guided the creation of clear and professional documentation. This demonstrates how AI, when used strategically, can accelerate learning, improve code quality, and support efficient problem-solving in software development.

Name: Modestah Inyanji Ngome
Course: AI